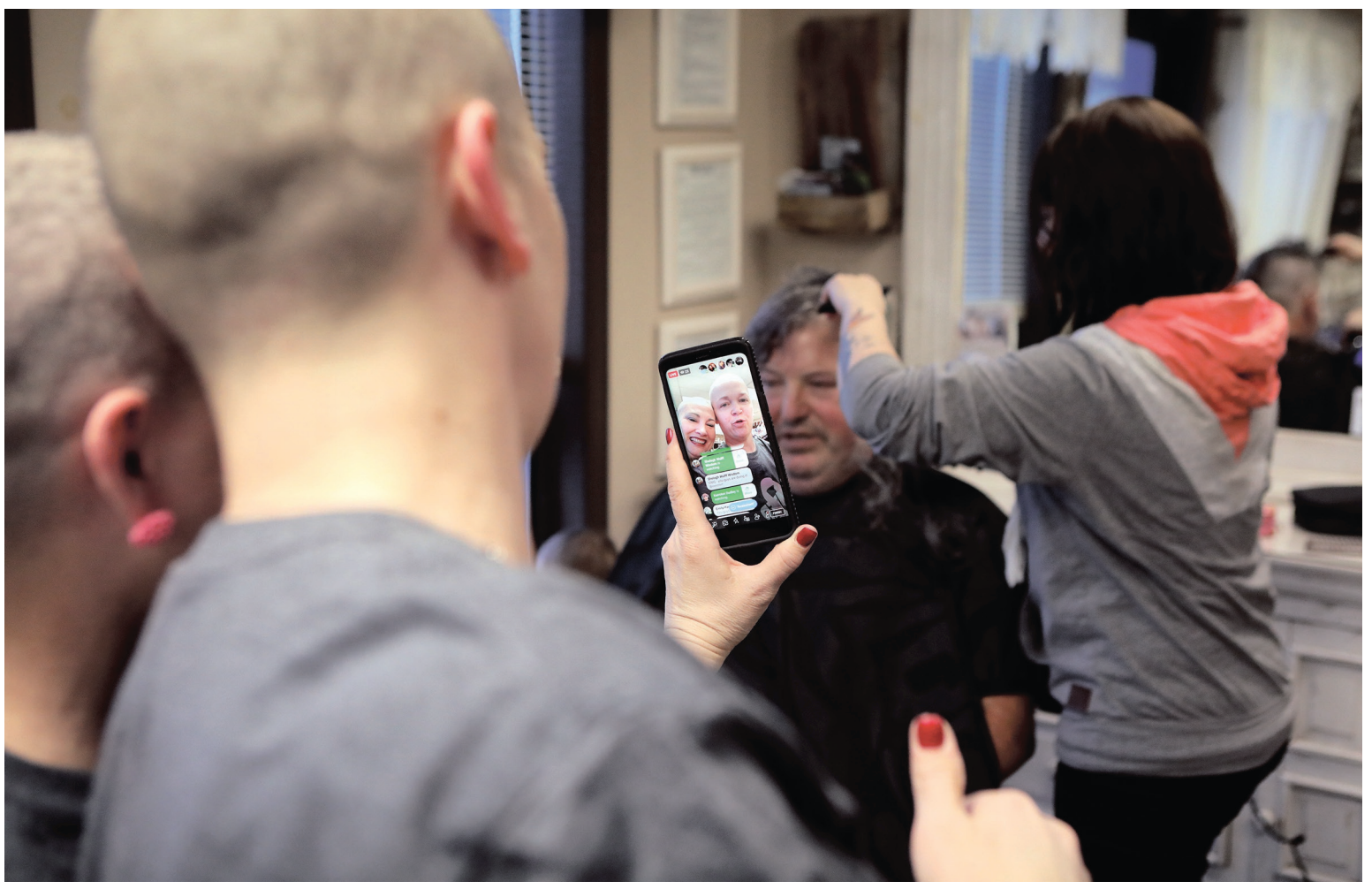
Douglas Budget

Vol. 133 No. 08 | Wednesday, February 20, 2019 | $1.00

Leaving it all on the floor



Mike Moore photo

Cathy Feezer-Jones (from left) and Jenni Nieft smile for the camera while live streaming a head-shaving fundraising event to support Douglas mayor Rene' Kemper as she begins a battle with aggressive breast cancer at Fringe Salon Feb 12. In the background, Desiree Wisecup starts the shaving process on previous mayor Bruce Jones' head.

Cadre of Rene' supporters sport bald tops

By Mary Stewart
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They came into the salon one by one. Some came with a spouse, some with a friend and some alone. This group had one purpose, one mission, and they were determined to see it through. If there was any example of people from all walks of life coming together to support a friend, this was it.

As each person came through the door, the level of excitement and nervousness grew and could be felt by everyone. It was like a bolt of lightening hit nearby and the buzz from the energy was floating in the air.

It was time to support Douglas' mayor, Rene' Kemper. It was time to shave some heads.

Kemper was diagnosed with stage four breast

Please see **GOING BALD,** page A-6

Will carbon capture be future for DJ plant?

By Colin Tiernan
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Picture the skeletal towers of oil derricks rising above the sage brush in 1919. The Big Muddy oil field practically oozed black gold, giving up a whopping 10,000 barrels a day. Roughnecks drove out to fields in Model Ts, ambitious fortune-seekers chaotically staked claims. It was Converse County's first boom, and Glenrock was a place of national interest and importance.



Today the Big Muddy field is nearly dry, producing less in a month than it mustered daily in the Roaring Twenties. A few rusty pump jacks sit forlornly here and there, even fewer of them still nodding up and down.

If the Dave Johnston power plant receives a much-touted carbon capture unit, dead oil fields for miles around could spring back to life thanks to enhanced oil recovery.

Fitting one of Dave Johnston's boilers with carbon capture technology and us-

Please see **CARBON CAPTURE,** page A-7

Wyo. coal has friends in the House

By Colin Tiernan
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Coal's decline might be inevitable, but it has friends in the Wyoming legislature.

Senate File 159 would require utilities to make an effort to sell power plants before shutting them down.

The bill has passed in the Senate with zero nay votes, and received unanimous approval from both Minerals Committees. It is now in the House.



REP. AARON CLAUSEN

Right now, public utilities can decommission coal-fired power plants without making an effort to keep them running.

Rep. Aaron Clausen (R-Converse) expects that the Dave Johnston power plant probably won't be the one benefiting from the bill. In the heart of a conventionally tapped-out oil field, Dave Johnston could very well face a different fate thant Sweetwater County's Jim Bridger power plant.

"I think Dave Johnston stands a bit of a better chance than some of the others," Clausen said. "If we can get some carbon capture for enhanced oil recovery, Dave Johnston will be in the driver's seat."

No Pain, No Gain



Mike Moore photos

Douglas Police Chief Ron Casalenda's watery breath is immediately carried away from his frigid body by strong winds following his splash at this year's Jackalope Jump Feb. 15. To watch a video from the jump this year, scan this image with your WyoInteractive app.

Jumpers break 10k goal at frigid fundraiser

By Mike Moore
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The temperature may have hovered in the mid 40s, but anyone who has lived in Douglas long enough knows that is not the only factor when taking wet winter risks in the month of February. The famed Wyoming winds howled as jumpers took the plunge into frigid waters last Friday, all with smiles on their faces as they knew the temporary pain was for the Wyoming Special Olympics' gain.

A hefty "pumpkin tank" used in the summer months to fight fire borrowed from the Bureau of Land Management was filled to the brim with ice-cold water in the vacant field behind Douglas High School. Being the month of February, the tank had a whole new use – to soak local fundraisers all in the name of Wyoming Special Olympics.

As eager youngsters huddled around the pumpkin-like structure, Jackalope Jump participants soon emerged in colorful, silly costumes to a round of applause.

Upon climbing a metal ladder up to the bed of an Army National Guard truck used as the platform for the plunge, some raised their arms high in celebration, while others covered their mouths in anticipation of frigid futures. And then, in a matter of seconds, they were wrapped in towels with teeth chattering away as the wind helped them dry off in a painful way.

Please see **JACKALOPE JUMP,** page A-3

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This week's weather

	Wed.	Thur.	Fri.	Sat.	Sun.
High	30	23	25	31	33
Low	7	9	8	14	13



Carbon capture

from the front page

ing that captured CO2 to revitalize tapped-out oil fields is no longer a new idea, but excitement remains high as several carbon capture plans are in the works here and nationwide

Two weeks ago, one of the companies vying for the carbon capture project at Dave Johnston, Glenrock Energy, talked with the Converse County commissioners about the technology and what it could mean for the county: Hundreds of jobs and millions of dollars in tax revenue.

Another similar company, Eco2Source, has been working in the background more quietly to gain approval from the owner of the Dave Johnston Plant, PacifiCorp.

As new legislation creates new incentives for carbon capture, PacifiCorp faces growing pressure to make a decision on the aging plant. An announcement on the future of carbon capture at Dave Johnston could come this summer.

HOW DOES IT WORK?

The most common carbon capture method is post-combustion capture. With post-combustion, you channel the effluent coal fumes, the flue stream of nitrogen, CO2 and water, through what amounts to a chemical net. Essentially, you force the CO2 through a chemical barrier, and those chemicals latch onto the CO2 molecules, dragging them out of the flue stream and creating a liquid out of a gas. The process, called amine scrubbing, pulls out 90 percent of the CO2 emitted by the boiler.

(There are actually three kinds of carbon capture, two of which are relevant at the Dave Johnston plant. The other kind is known as oxy-combustion capture, where you burn coal in a pure oxygen environment, which simplifies the CO2 capture process. In theory, Dave Johnston could have both, on different boilers, at the same time.)

After grabbing the CO2 out of the flue stream, you're left with a non-pure CO2 liquid. Extracting the C02 from that liquid isn't cheap. In order to break the bond between the chemical solvent and the CO2, you need to superheat the liquid to about 200 or 300 degrees Fahrenheit. That ends up requiring massive amounts of energy, about a third of all the energy produced by the boiler.

Director of Research at the University of Wyoming School of Energy Resources Scott Quillinan explained the demand of carbon capture technology on a boiler.

"It adds a parasitic load to the power plant," Quillinan said. "At the end of the day, carbon capture costs money, so you have to be able to use it (CO2) somewhere to make up for those economics."

That parasitic load is part of the reason why aging power plants, that are nearing the ends of their useful lives, make the best carbon capture candidates: The energy they produce can potentially become less crucial as part of the overall power grid and dedicating the whopping 33 percent of produced energy to carbon capture becomes more palatable.

THE MONEY MAKER

Carbon capture sounds like a good idea from an environmental standpoint. But environmental ethics aren't the primary driver of the excitement surrounding a potential Dave Johnston carbon capture unit (in fact, while environmentalists tout sequestration, many point out that tertiary recovery doesn't reduce CO2 emissions nearly as much). Captured carbon can be worth big money because it can resuscitate dead oil fields.

A hundred years ago, Wyoming oilmen approached drilling the same way you approach a milkshake. They'd drill a hole and pump crude with the help of the natural pressure built up in the reserve. As they drilled, pressure in the underground reserve dissipated, and the field effectively ran dry, even though huge amounts of oil, about 90 percent, remained.

A few decades ago, Texas operators discovered an advantageous way of hydraulically fracturing rock. Shoot some water into a dry field at high pressure, move the oil around, open up some cracks, and formerly trapped oil can seep through. Fracking allows for massive amounts of new drilling in previously inaccessible areas, and new improvements have enabled operators to drill horizontally, too. But eventually, even fracking has its limits. You've gotten to oil you had no chance of getting to in the 1940s, but there's still a huge amount (this time 80 to 60 percent) of crude lingering at the bottom, enticingly.

Enter tertiary recovery, or enhanced oil recovery. When you inject captured CO2 into an oil reserve, the CO2 molecules interact with the oil.

"When you capture it, compress it, and send it down in an oil reservoir, it's a supercritical fluid," Quillinan said. "It actually works as a solvent. Oil and water don't mix; CO2 and oil do."

This reaction causes the oil to expand, and when it expands it moves. You're altering the nature of the oil and making it more susceptible to extraction. You're pushing the oil toward your bore, and essentially lubricating the crude by chemically reducing its viscosity.

IF IT'S SO GREAT, WHY ISN'T EVERYONE DOING IT?

Carbon capture could expand rapidly in the coming years, but because of the costs and uncertainties of the technology, few power plants have taken the plunge.

There are two active carbon capture facilities roughly analogous to what Dave Johnston would become. There's the Petra Nova plant outside of Houston, and SaskPower's Boundary Dam plant in Canada just across the North Dakota border.

Petra Nova needed government help to get its project off the ground, in the form of a $190 million grant. SaskPower took $240 million from the Canadian government. Both of those plants send their CO2 to nearby oilfields for tertiary recovery, and both are pretty new. Petra Nova has only been operational for two years.

The technology isn't cheap, either.

One estimate states that a Dave Johnston carbon capture system would require upwards of $800 million to construct.

At the meeting with the Converse County commissioners, Glenrock Energy's Terry Manning estimated the carbon capture unit itself would cost $450 million, and the enhanced recovery system at the oil fields and the pipeline would come out to $250 million. Manning also said the project would bring $750-800 million to the state and county.

If the Dave Johnston carbon capture project were to become a reality it, too, would rely on heavy government subsidization, like Boundary Dam and Petra Nova did. With tax credits, Dave Johnston carbon capture could be helped to the tune of $100 million annually.

Recent legislation has excited carbon capture hopefuls. The FUTURE Act (the acronym is a bit of a stretch, it stands for Furthering carbon capture Utilization, Technology, Underground storage and Reduced Emissions) triples the value of carbon capture tax credits by amending the IRS tax code's section 45Q.

Before the FUTURE Act, the government incentivized carbon capture with a credit of $10 per ton. That number would now be a bit over $33 over 12 years at Dave Johnston (the number is $50 if you're purely sequestering it). The act nixed the former 75,000 ton CO2 cap as well.

Eco2Source, one of the companies vying for the Dave Johnston project, estimates CO2 capture at the plant could be worth as much as $100 million annually in tax credits. That company's owners declined to comment on the record about their project.

The FUTURE Act isn't open-ended, though. If a company wants to reap the lucrative tax credit rewards, it has to break ground on its project by January 2024, and the credits only last for 12 years following commencement of operations. If the Dave Johnston project became a reality, the tax credits alone could be worth $1.2 billion.

Glenrock and Converse County would benefit because of jobs created in formerly tapped out, or meagerly producing, oil fields. Manning estimates 1,200 to 1,500 total jobs, including the retention of Dave Johnston jobs, would be created as a result of the project.

PacifiCorp's role as a public utility throws a wrench into potential carbon capture projects, too. Public utilities don't have the same prerogatives as purely private companies. PacifiCorp has to ensure any alterations to the Dave Johnston facility are either neutral or positive for consumers and don't lead to rate hikes. If a carbon capture unit required the plant to raise rates, it'd be a non-starter.

PERFECTLY POSITIONED

The Petra Nova plant has to pipe its CO2 80 miles before it can be used for enhanced recovery.

At Dave Johnston, the pipes would only have to stretch a few miles before arriving at oil fields. Big Muddy, Teapot Dome and Salt Creek would be the primary candidates. In the region, there are well over a hundred million barrels of oil that could be reached with enhanced recovery, experts say. Manning, whose company owns the Big Muddy field, estimates tertiary recovery would lead to a 25-year renewed life. He added that tertiary recovery in nearby fields would have a low $29 break-even point.

Those in the carbon capture industry say that pressure for carbon capture is building. The technology has strong political support. Gov. Mark Gordon has been vocal about his support for carbon capture, as was former Gov. Matt Mead. Sen. John Barrasso (R-Wyo) was a co-sponsor of the FUTURE Act, and Converse County politicians all tout the technology as invaluable to the region.

Local representatives point to the potential for saving jobs at Dave Johnston and creating hundreds more by expanding production in Big Muddy, Teapot Dome and Salt Creek. A recent state legislative bill would require power plant owners to look for a buyer before shutting plants down.

If PacifiCorp decides to pass on a carbon capture unit, and continues with its plans of retiring Dave Johnston in 2027, a new owner could buy the plant, possibly with the specific intention of implementing carbon capture.

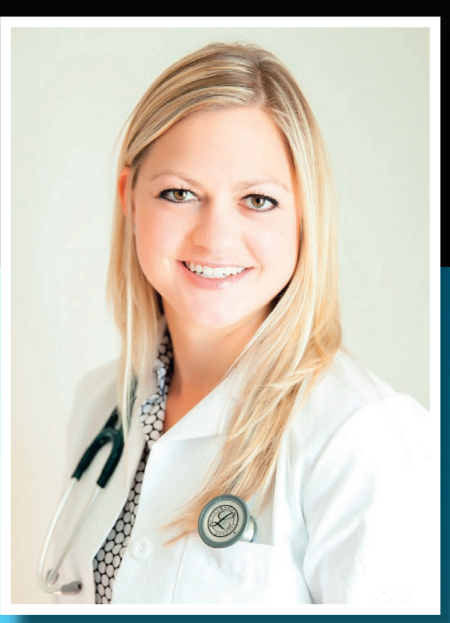
The county commissioners expressed their receptiveness to putting a carbon capture system in place at Dave Johnston.

"We are very much supportive of the general premise," Converse County Commission Chairman Robert Short said. "Obviously from an economics driver, this is only sensible. It's far more than just tertiary oil recovery, it extends life to 183 employees currently at D.J., as well as, ancillarily speaking, it plays into direct production of coal."

PacifiCorp's decision could be coming soon. The company has sent out a call for carbon capture project proposals, due by the end of February. After that, the company is likely to conduct a feasibility study. Manning believes a final decision could be made late this summer, in part because beyond August 2019, it will be difficult for a company to meet the FUTURE Act's 2024 deadline.

"We need to push this along sooner rather than later," Manning said in an interview. "Every month or two or three that gets chewed up in this evaluation phase is time you're taking away from the three year plus lead time you need to get the project in place."

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